Ian John

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EDUCATION

Georgia Institute of Technology, Atlanta, GA — Expected May 2028
Bachelor of Science in Computer Engineering
John H. Martinson Honors Program
GPA: 4.0

University of Georgia, Athens, GA — Aug 2025 - May 2026
Computer and Mechanical Engineering
GPA: 3.97

Augusta University, Augusta, GA — Aug 2024 - May 2025
Dual Enrollment Coursework
GPA: 4.0

EXPERIENCE

Waller Home Solutions, *Property Maintenance Technician*, Augusta, GA — Jul 2025 - Aug 2025
- Serviced commercial and residential properties through detailed cleaning, apartment turnover preparation, pressure washing, landscaping, and exterior maintenance
- Delivered reliable, high-quality results in physically demanding job-site environments while maintaining strong attention to detail and client property standards

Self-Employed, *Independent Math Tutor*, Evans, GA — Jan 2025 - May 2025
- Tutored a high school student in upper-level math through customized explanations and guided problem-solving
- Created practice materials to strengthen algebraic understanding, homework performance, and test confidence

INVOLVEMENT

UGA Robotics Club - BattleBots Division, *Mechanical/Electrical Team Member*, Athens, GA — Sep 2025 - May 2026
- Designed, fabricated, and tested competitive combat robots in collaboration with a multidisciplinary team
- Supported mechanical design, electrical integration, troubleshooting, and pre-competition testing
- Concepted and built a first-of-its-kind electronic drive system from the ground up

PROJECTS

NAND-Only Digital Calculator, *Discrete Logic Design - In Development*, Atlanta, GA — Aug 2026 - Present
- Architecting a digital calculator from 2-input NAND gates, translating arithmetic and control functions into gate-level logic
- Planned IC/gate counts, breadboard layout, wiring, power distribution, decoupling capacitors, BOM, and subsystem testing

Minecraft Digital Logic Systems, *Independent Computer Engineering Project*, Atlanta, GA — Jul 2026
- Designed and built fully functional digital logic systems in Minecraft Redstone using only logic gates constructed from scratch, without command blocks, prebuilt components, or gameplay-based shortcuts
- Derived and tested a 3-to-8 decoder, 8-to-3 encoder, 8-to-1 multiplexer, 3-bit magnitude comparator, fixed +5 arithmetic circuit, SR latch, and 4-bit register using truth tables and Boolean expressions

Independent Janssen Effect Project, *Experimental Physics / Instrumentation*, Atlanta, GA — May 2026
- Designed and built a low-cost apparatus to investigate force redistribution in granular materials through the Janssen effect
- Used an Arduino Uno, HX711 amplifier, and load cell to measure force transmitted to a suspended bottom plate as rice was added incrementally
- Calibrated the sensor system and debugged the force path, showing how wall friction and granular force chains reduced bottom-load transmission

Reverse-Engineered Digital Watch CAD Assembly, *Mechanical Design / Autodesk Inventor*, Athens, GA — Apr 2026
- Reverse-engineered a consumer digital watch by disassembling, measuring, and recreating its components as an eight-part Autodesk Inventor assembly
- Produced detailed part models, dimensioned drawings, a bill of materials, exploded assembly documentation, and orthographic, section, and isometric views

SKILLS

Design & Manufacturing: Autodesk Inventor, CAD modeling, dimensioned drawings, bill of materials, 3D printing, prototyping, fabrication
Test & Instrumentation: Load cells, ultrasonic sensors, multimeter, sensor calibration, experimental data collection
Hardware: Arduino, breadboarding, soldering, SMD soldering, circuit construction, sensor integration, electronics repair
Programming: Java, MATLAB, Python (familiar)
Development Tools: Arduino IDE, Microsoft Word, Excel, PowerPoint
Engineering: Mechanical/electrical integration, troubleshooting, technical documentation
Honors: National Merit Finalist